KIRLOSKAR CHILLERS PRIVATE LIMITED
A Kirloskar Group Company

ENGINEERING SPECIFICATION

Document Name: Chiller Design and Test Pressures | **Document No.:** ESP/07-08/107

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Chiller	Model	Refrigerant		Heat Exchanger Pressures									Unit Pressures	
				Maximum Operating (Refrigerant side)	Design (Refrigerant side)	Maximum Operating (Water side)	Design (Water side)	Tube Side Test [4] Pressure		Shell Side [4 & 5] Test Pressure		Safety valve		Unit test Pressure [1] without relief valve (Refrigerant side)
				Bar	Bar	Bar	Bar	kgf/cm² Hydrostatic	kgf/cm² Pneumatic	kgf/cm² Hydrostatic	kgf/cm² Pneumatic	Bar		kgf/cm² Pneumatic
WATER COOLED	KWS	R22/R407C	Evaporator	14.5	16.0	9.4	10.3	NA	17.9	13.7	NA	16.0	LP	17.9
			Condenser	19.1	21.0			13.7	NA	NA	23.6	21.0	HP	23.6
		R134a	Evaporator	11.8	13.0	9.4	10.3	NA	14.6	13.7	NA	13.0	LP	14.6
			Condenser	14.5	16.0			13.7	NA	NA	17.9	16.0	HP	17.9
		R1234Ze	Evaporator	9.4	10.3	9.4	10.3	NA	11.6	13.7	11.6	10.3	LP	11.6
			Condenser	11.8	13.0			13.7	NA	NA	14.6	13.0	HP	14.6
	KWS Brine	R404A	Evaporator	19.1	21.0	9.4	10.3	NA	23.6	13.7	11.6	21.0	LP	23.6
			Condenser	21.8	24.0			13.7	NA	NA	26.9	24.0	HP	26.9
	KWK/KWI/KWE/ KSC/KDC/KSM/KCM	R134a	Evaporator	11.8	13.0	9.4	10.3	13.7	NA	NA	14.6	13.0	LP	14.6
			Condenser	11.8	13.0						14.6	13.0	HP	14.6
	CE marked - KWK/KWI/KCM	R134a / R1234Ze	Evaporator	11.8	13.0	9.4	10.3	15.0	NA	NA	19.0	13.0	LP	19.0
			Condenser	14.5	16.0						23.3	16.0	HP	23.3
	Unit with PED vessels only KWK/KWI/KCM	R134a	Evaporator	11.8	13.0	9.4	10.3	15.0	NA	NA	19.0	13.0	LP	19.0
			Condenser	14.5	16.0						23.3	16.0	HP	23.3
	CE marked - KSC/KDC	R134a	Evaporator	9.6	10.6	9.4	10.3	15.0	NA	NA	15.5	10.3	LP	15.5
			Condenser	14.5	16.0						23.3	16.0	HP	23.3
	Unit with PED vessels only - KSC/KDC/KSM	R134a	Evaporator	9.6	10.6	9.4	10.3	15.0	NA	NA	15.5	10.3	LP	15.5
			Condenser	14.5	16.0						22.9	16.0	HP	23.3
	KWI	R1234ze	Evaporator	9.4	10.3	9.4	10.3	13.7	NA	NA	11.6	10.3	LP	11.6
			Condenser	11.8	13.0						14.6	13.0	HP	14.6
AIR COOLED	KAS/KAA	R22/R407C	Evaporator	14.5	16.0	9.4	10.3	NA	17.9	13.7	NA	16.0	LP	17.9
			Condenser	25.5	28.0	NA	NA	NA	36.7	NA	NA	28.0	HP	31.4
		R134a	Evaporator	11.8	13.0	9.4	10.3	NA	14.6	13.7	NA	13.0	LP	14.6
			Condenser	19.1	21.0	NA	NA	NA	36.7	NA	NA	21.0	HP	23.6
	KAC	R410A	Evaporator	25.0	27.5	9.4	10.3	NA	31.0	10.5	NA	27.5	LP	31.0
			Condenser	37.7	41.5	NA	NA	NA	46.5	NA	NA	41.5	HP	46.5

Notes:

1. Unit test pressure with Relief valve = 0.8 × Design Pressure (Safety valve set presure). For LP side consider evaporator safety valve set pressure and for HP side consider condenser safety valve set pressure.

2. For Oil seperator pressures, Refer respective Condenser pressures and safety valve rating.

3. Design temperature for Standard & CE (except CE Centrifugal) Evaporator & Condenser is 65°C & 100 °C Respectively. Design temperature for CE Marked Centrifugal Evaporator & Condenser is 65 °C & 65 °C Respectively.

4. All the pressures mentioned above are applicable for standard chillers. Don't refer these in case of Heat Pumps.

5. On Refrigerent side, hydrostatic test shall not be performed. According to ASME Sec VIII Div I, UG-100, for vessels where water traces are harmful, hydrotest can be avoided. Only pneumatic test is sufficient.

6. To identify Unleash Heat exchanger refer PRV added below top code.